**Name:** _____

**Class:** _____

**Mr. Rawson • GCSE Computer Science** Year 11 2026

# Algorithms 2 — Flowcharts and Trace Tables

## 1 The six flowchart symbols

**VOCAB** **Flowchart** (OCR 2.1.2) — a diagram of an algorithm, drawn with standard symbols joined by arrows that show the order of the steps

A flowchart shows the same steps as a program, as a picture. OCR uses six symbols. In the exam you may be asked to read a flowchart, complete one, or draw your own, and a Section B design answer can be given as a flowchart.

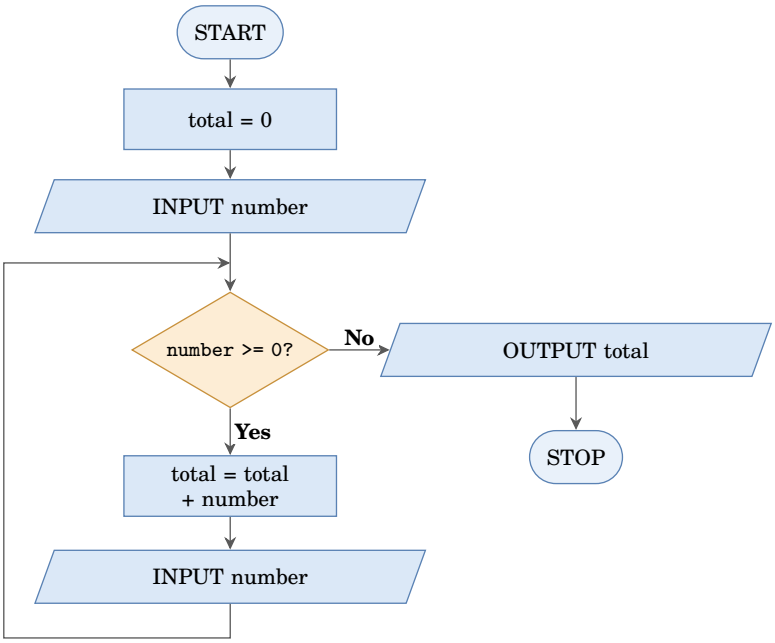
| Symbol | Name | Use it for |
|---|---|---|
| START | Terminal | The start and the end. Every flowchart begins with START and finishes with STOP. |
| INPUT number | Input / output | Data coming in from the user, or going out to the screen. |
| total = 0 | Process | A step that does something: a calculation, or giving a variable a value. |
| age > 17? | Decision | A question with two answers. It has two exits, labelled **Yes** and **No**. |
| show_menu() | Sub program | Running a sub program that is defined somewhere else. |
| → | Line | Joins the symbols. The arrow shows which step comes next. |

## 2 A flowchart beside its program

**The rule.** Start at START and follow the arrows. At a decision, answer the question and leave by the exit with that label. A **loop** is an arrow that goes back up to a decision, so the steps below it run again.

**Worked example: Sum of Positive Numbers**, from your holiday sheet.

```
1 total = 0
2 number = int(input("Enter a number: "))
3 while number >= 0:
4     total = total + number
5     number = int(input("Enter a number: "))
6 print("The sum of positive numbers is:", total)
```




| Shape | Line of the program |
|---|---|
| Process | 1: `total = 0` |
| Input | 2 and 5: `int(input(...))` |
| Decision | 3: the `while` condition |
| Process | 4: add to the total |
| Output | 6: `print(...)` |

The arrow from the second INPUT goes back up to the decision: that is the `while` loop. The decision comes **before** the steps it repeats, exactly as the `while` line comes before its indented lines.

**Now try** • Draw the flowchart for your **Even or Odd** holiday program. Use `MOD`, the exam's word for %: the decision is `number MOD 2 == 0`?

**Check** • One START, one STOP, one decision with both exits labelled, and an output on each exit: `Even` on the Yes side, `Odd` on the No side. Follow it with 7 and you reach `Odd`.

## 3 Trace tables

**VOCAB** **Trace table** (OCR 2.1.2) — a table used to follow an algorithm one line at a time, recording the value of each variable, to check what it does and to find logic errors

**The rule.** Give each variable its own column, plus a column for the condition and one for the output. Go through the program one line at a time, in the order the computer runs them. **When a line gives a variable a value, write that value in the variable's column on a new row.** When a line prints, write what it prints in the Output column.

**Worked example.** The Sum of Positive Numbers program above, with the inputs **5**, **12** and **−1**.

| Line | number | number >= 0 | total | Output |
|---|---|---|---|---|
| 1 | | | 0 | |
| 2 | 5 | | | |
| 3 | | True | | |
| 4 | | | 5 | |
| 5 | 12 | | | |
| 3 | | True | | |
| 4 | | | 17 | |
| 5 | −1 | | | |
| 3 | | False | | |
| 6 | | | | The sum of positive numbers is: 17 |

Lines 3, 4 and 5 repeat, once for each positive number. The loop ends when the condition is **False**, and only then does line 6 run. The last value in the total column is the value that gets printed.

**Now try** • Trace the same program with the inputs **8**, **0**, **3** and **−2**.

**Check** • The output is `The sum of positive numbers is: 11`. The condition column reads True three times, then False once: 0 is *0 or more*, so it goes round the loop.

| Line | number | number >= 0 | total | Output |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |

## 4 The exam writes it in ERL

Exam trace tables are usually set on OCR Exam Reference Language (ERL). It is the same program: ERL closes the loop with `endwhile` where Python uses the indent, and joins text with +.

**PYTHON (what you type)**

```
total = 0
number = int(input("Enter a number: "))
while number >= 0:
    total = total + number
    number = int(input("Enter a number: "))
print("The sum of positive numbers is:", total)
```

**OCR ERL (what the exam shows)**

```
total = 0
number = int(input("Enter a number: "))
while number >= 0
    total = total + number
    number = int(input("Enter a number: "))
endwhile
print("The sum of positive numbers is: " + str(total))
```

**A counting loop in ERL.** `for i = 1 to 3` gives `i` the values 1, 2 and 3, **including** the 3. It is Python's `for i in range(1, 4):`, and `next i` closes the loop.

**Now try** • Trace this ERL program. It is your Multiplication Table holiday program for the number 7, cut to three lines of the table.

```
number = 7
for i = 1 to 3
    answer = number * i
    print(answer)
next i
```

**Check** • The Output column reads 7, 14, 21, and `i` takes the values 1, 2 and 3: the first three lines of the 7 times table.

| Line | number | i | answer | Output |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |

## Stretch  Guess the Number as a flowchart

Draw the flowchart for your **Guess the Number** holiday program. The `while` loop is a decision with the arrow coming back up to it, as in Section 2. Inside the loop, the `if`, `elif` and `else` become two decisions, one below the other.

**Check** • One START and one STOP, three decisions (`guess != secret?`, `guess > secret?`, `guess < secret?`), and every output inside the loop has an arrow back up to the first decision.

## The exam answers, one line each

| Asked for | Write |
|---|---|
| Flowchart | A diagram of an algorithm using standard symbols joined by arrows. |
| Decision symbol | A diamond, for a question with two exits labelled Yes and No. |
| Trace table | A table that records the value of each variable as an algorithm runs, line by line. |
| Why use a trace table | To check what an algorithm outputs, and to find logic errors. |